$$Q=\frac{7}{2}nR\left(T_a-T_1\right),\tag{2}$$

This heat equals the latent heat absorbed as the iceberg melts, so

$$Q=mL,\tag{3}$$

Combining (2)、(3) gives

$$n=\frac{2mL}{7R\left(T_a-T_1\right)}。\tag{4}$$

In isochoric heating step (b), let the final pressure be p_0 and volume V_0. Then

$$p_a=\frac{T_a p_a}{T_1},\tag{5}$$

$$V_0=\frac{nRT_a}{p_0}\tag{6}$$

In each small isothermal jet step, the vessel pressure changes by Δp, with

$$\frac{p+\Delta p}{p}=\frac{V_0-u}{V_0},\tag{7}$$

Here V_0 is the volume in step (b), and p the vessel pressure. The expelled gas has kinetic energy

$$\Delta E=\left(p-p_a\right)u,\tag{8}$$

Eliminating u using (7) gives

$$\Delta E_k=-\left(p-p_a\right)V_0\frac{\Delta p}{p}。\tag{9}$$

Taking $\Delta E\to dE$, $\Delta p\to dp$, the total kinetic energy is

$$E_{\mathrm{k}}=\int_{p_0}^{p_a}-\left(p-p_a\right)V_0\frac{dp}{p}\tag{10}$$

$$=\left(p_0-p_a\right)V_0-p_aV_0\ln\frac{p_0}{p_a}$$

The total electrical energy obtained under the given conditions is

$$E=0.45E_{\mathrm{k}},\tag{11}$$

Substituting (11)(5)、(6)、(10) into gives

$$E=\frac{9}{70}mL\frac{1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}}{1-\frac{T_1}{T_a}};\tag{12}$$

Substituting the numerical data gives

$$E=1.5\times10^{14}\ \mathrm{J}。\tag{13}$$